

JOSEPH AMARA

MSc in Geo-information Science Student

Geoscience | Environmental Assessment | GIS & Remote Sensing | Environmental Monitoring

Wageningen, Netherlands • Sierra Leonean

Email: josephamara3284@gmail.com, joseph.amara@wur.nl

Phone: +232 76 456149 | +31 684459134

LinkedIn: [Joseph Amara](#)

Languages: English (fluent)

Geo-visualisation Portfolio: [Online Portfolio](#)



PROFESSIONAL PROFILE

I am an Environmental and GIS professional with a First-Class BSc (Honours) in Geology from Fourah Bay College, University of Sierra Leone, and currently completing an MSc in Geo-Information Science at Wageningen University & Research. Professional experience covers environmental and social impact assessment, environmental monitoring, spatial analysis, field data collection, environmental reporting and support to mining, power and solar projects in Sierra Leone.

My background combines field experience with geoscience training and growing expertise in GIS, remote sensing, spatial modelling and data analysis. I am experienced in working in multidisciplinary teams, field supervision, environmental datasets and turning field and spatial evidence into clear findings for clients and regulatory stakeholders. I am most interested in using geospatial evidence to support sound environmental management, responsible natural resource use and sustainable development.

CORE AREAS OF EXPERTISE

- Environmental & Social Impact Assessment
- Environmental Monitoring & Compliance
- Geoscience & Natural Resources
- Field Survey & Data Collection
- Spatial Modelling & Statistics
- Stakeholder & Field Team Support
- GIS & Spatial Analysis
- Remote Sensing & Earth Observation
- Environmental Data Management
- Environmental Reporting & Technical Writing
- Water, Air & Noise Monitoring
- Mining, Energy & Development Projects

PROFESSIONAL EXPERIENCE

Environmental and GIS Consultant | Integrated Geo-information and Environmental Management Services (INTEGEMS) Ltd., Freetown, Sierra Leone

April 2023 – August 2025

- Conducted Environmental and Social Impact Assessments (ESIA), supporting clients from EIA licensing through environmental compliance with Sierra Leonean requirements, IFC standards and Good International Industry Practice.
- Supervised routine environmental monitoring and assessment for two large-scale mining companies, including field planning, data collection, quality control and reporting.
- Collected and analysed spatial and temporal data from field and satellite sources to support environmental monitoring, site selection and assessment.
- Conducted field monitoring of water quality and quantity, air quality and noise using calibrated instruments and established monitoring procedures.
- Processed and interpreted environmental monitoring data and prepared technical reports with findings and practical recommendations for clients and regulatory bodies.
- Developed multi-layered maps and conducted spatial analysis using ArcGIS Pro and other GIS tools to support environmental assessment and project planning.
- Managed geodatabases containing environmental and geological information, maintaining data quality and accessibility for project teams.

- Supervised a team of three field technicians and coordinated field activities to meet environmental monitoring and reporting requirements.
- Provided geological and environmental input to projects and advised clients on environmental and social issues and occupational health and safety.

Intern Geologist | National Minerals Agency, Freetown, Sierra Leone

April – August 2022

- Used GIS to identify suitable stream sediment sampling points by integrating hydrological and topographic information.
- Supported geological fieldwork, sample preparation, rock and core description and mineralogical interpretation.
- Participated in site visits to large-scale mining operations and supported audits of exploration companies against national mining and environmental requirements.
- Applied geological and spatial evidence to support mineral exploration and regulatory reporting.

Research Assistant | Economic and Environmental History Group, Wageningen University and Research

Officially starts September 2026.

- Georeference historical maps
- Create training samples for the training of Machine Learning and Deep Learning Models.
- Clean and process large spatial datasets.
- Generate outputs from trained ML and DL models.

SELECTED PROFESSIONAL PROJECT EXPERIENCE

Routine Environmental Monitoring and Assessment – Gold Lion Mining Company (GLMC) and CTC Mining (SL) Limited

Environmental and social monitoring for two large-scale mining operations in Sierra Leone; served as team lead during the final year of the assignment.

Environmental Monitoring – CECA-SL 108.4 MW LPD/LNG Nant Power Generation Project

Supported and led environmental monitoring during construction of one of Sierra Leone's major recent power generation projects.

Environmental and Social Impact Assessments – Mining, Exploration and Solar Projects

Contributed to ESIA's for mining, exploration and solar power projects, including geological, environmental and spatial assessment tasks.

Geological and Structural Mapping – North-eastern Sierra Leone

Undergraduate research applying high-resolution aeromagnetic and remote sensing data to geological and structural interpretation.

Groundwater Quality and Contamination Research – Sierra Leone

Research contributing to published studies on heavy metal contamination and nitrate sources in groundwater around major dumpsites.

EDUCATION

MSc Geo-Information Science | Wageningen University & Research, Netherlands | September 2025 – Present

Relevant coursework with grades: Remote Sensing (9.0); Advanced Earth Observation (8.5); Geoscripting (8.0); GIS in Context (8.5); Geo-information Tools (8.0); Machine Learning (8.0); Spatial Modelling and Statistics (7.0); Planning and Design of Urban Space (8.5), Remote Sensing and GIS Integration (8.5).

BSc (Honours) Geology – First Class, GPA 3.93/4.00 | Fourah Bay College, University of Sierra Leone | 2018 – 2022

Relevant areas: Hydrogeology, GIS and Remote Sensing, Geochemistry, Geophysics, Structural Geology, Petrology, Petrogenesis, Environmental Geology and Petroleum Geology.

TECHNICAL SKILLS

GIS

ArcGIS Pro, QGIS, Google Earth Pro

Remote Sensing

LiDAR, Sentinel-1&2, LandSAT, MODIS

Programming

Python, R

Geoscience software

RockWorks, Geosoft Oasis Montaj, Surfer

Office & reporting

Microsoft Word, Excel, PowerPoint, Visio, Outlook

Data Visualisation

Tableau

Field Monitoring

Water, air and noise quality; geological, environmental and social field surveys and stakeholder engagements.

ACADEMIC PROJECTS & RESEARCH

- Detailed description of Master's academic projects can be found in my [personal portfolio](#).
- Undergraduate Thesis (August 2022 to December 2022): Geological and Structural Mapping of Parts of North-Eastern Sierra Leone using Integrated Aeromagnetic and Remote Sensing Data.
- Master's Thesis (September 2026 to February 2026): Causal Floods in Sub-Saharan Africa.
- Use of Terrestrial, Mobile and Airborne LiDAR scanners during my Remote Sensing and GIS Integration fieldwork. Processed LiDAR data during my Remote Sensing and Advance Earth Observation Courses for terrain and vegetation analysis.

SCHOLARSHIPS, AWARDS & HONOURS

- Wageningen University & Research African Scholarship Programme Scholar, 2025–present.
- Adonis Abboud Trust Fund Award – Best Overall Graduating Student, Fourah Bay College, University of Sierra Leone, Class of 2022 (GPA 3.93/4.00).
- National Education Champions Award – Best Student, Fourah Bay College, 2021/2022 academic year.
- Academy Excellence Award, Fourah Bay College Student Union Government.
- Wageningen University Master's Honours Programme Student

PROFESSIONAL and ACADEMIC MEMBERSHIP

Sierra Leone Institution of Geo-Scientists (SLIG) – Member | 2020 – Present

United Community of African Students, Wageningen University & Research – Vice President | June 2026 – Present

TRAINING & SHORT COURSES

- Esri – Getting Started with Spatial Analysis, Getting Started with Data Management and GIS Basics.
- Centre for Exploration Targeting, The University of Western Australia – Mineralogy and Geochemistry of Gold Ore.
- DataCamp – R (Intro and Intermediate courses), Python (Intro and Intermediate courses) and Bash (Intro course).

PUBLICATIONS

- Sankoh, A. A., Amara, J., Komba, T., Laar, C., Sesay, A., Derkyi, N. S., & Frazer-Williams, R. (2023). Seasonal assessment of heavy metal contamination of groundwater in two major dumpsites in Sierra Leone. Cogent Engineering, 10(1).
- Sankoh, A. A., Derkyi, N. S. A., Laar, C., Edegbene, A. O., Asomaning, J., Amara, J. & Frazer-Williams, R. (2025). Using a Bayesian isotope mixing model to assess nitrate sources in groundwater: A case study of Granvillebrook and Kingtom dumpsites, Sierra Leone. Journal of African Earth Sciences, 226, 105585.

HOBBIES

Playing football, reading, watching movies and tv shows, YouTube (Science, Linguistics and Educational channels).

REFERENCES

Professional references are available on request. Named academic and professional referees can be provided for formal applications where required.